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Skills

Generative AI: Prompt Engineering, Hugging Face, Transformers
Deployment, Containerisation
Docker, TailScale, SkyPilot, Flask Streamlit,
DBMS:
MySQL, PostgreSQL, MongoDB, MariaDB
Analytics & Visualisation:
Python, Power BI, MS-Excel
Data Science Libraries:
Pytorch, Keras, sklearn, pandas,
Programming: Data Versioning, HTML,CSS,

Certifications

Google IT Support Certification,
Introduction to Linux,
SQL for Data Science,
Machine Learning Foundations: A Case Study Approach ,
Crash Course on Python

Teaching

Teaching Assistant, NPTEL
(Jan 2024- April 2024)
noc24-cs47:

The Joy of Computing Using Python
Prepared weekly assignments & handled the forum for the course
having around 60,000 registered learners.

Student Mentor, IITM BS Degree Programme
(Mar 2024- April 2024)
BSMA1004:

Statistics for Data Science 2.
Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of 25 students

Maria Paul T

ASPIRING DATA SCIENTIST

Summary

Master's Degree Student building skills in the field of Data Science, Passionate about research and technology. Looking for opportunities to integrate research with latest technologies.

Experience

Research Intern, IIT Jammu | June 2026- Present

Quality-Clustering

- Developed statistical validation framework for quality-bucketing of large text corpora. Implemented topic erasure via linear residualization (confounding control); validated with before/after hypothesis testing (linear probing). Used bootstrap Jaccard resampling (20 iterations) to assess cluster stability;

Gaperon-llm-as-a-judge

- Built automated evaluation framework using LLMs to assess text quality across 6 dimensions (clarity, coherence, depth, grammar, usefulness, overall).
- Implemented concurrent batch processing for efficient data handling; compared open-source (Llama 3.2) vs. proprietary (OpenAI) judge performance via API abstraction.
- Validated judge reliability (inter-rater agreement surrogate) and identified systematic biases in scoring patterns.

Multilingual LLM Evaluation Pipeline

- Built Kubernetes-based evaluation infrastructure (V100 GPU cluster, SkyPilot orchestration) for large-scale multilingual LLM assessment. Debugged backend resource allocation and model-serving issues; implemented distributed evaluation across 4,900+ samples. Generated reproducible artifact pipeline: technical reports + interactive HTML visualizations with embedded plots and sample-level data.

Gaperon mmbert distill repository

- Engineered end-to-end ML pipeline (HuggingFace Hub integration, SkyPilot orchestration) for multilingual model distillation: mmBERT-base → 6-language classifier with independent heads (clarity, coherence, depth, grammar, usefulness, overall). Implemented dataset streaming, checkpoint versioning, and model promotion across branches; optimized for cost and latency across 6 languages (English, Hindi, Tamil, Bengali, Telugu, Malayalam).

Synthia - Synthetic Data Generation Pipeline

- Built production-grade synthetic instruction-tuning data pipeline: raw text → structured triples (context, instruction, response) via self-hosted LLM. Integrated with a private GPU cluster (Tailscale VPN); implemented language-normalized quality filtering (98.9% validation rate) with audit logging.
- Supports pure and code-mixed outputs (Hinglish/Tanglish) with downstream quality tracking—trace any training sample to its generation logic and parameters.

Summer Intern, S3 Labs, IIT Bhilai | May 2025 - July 2025

1. Technical Image Understanding

- Evaluating how Packet Diagrams, Sequence Diagrams, Class Diagrams, State Diagrams, etc are inferred using LLM's..

2. Legal Infographic Generation

- Dataset : IL-TUR(Exploration Lab, IIT Kanpur)
- Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain
- Generated trial prototypes for generating Infographics like Timelines, Key Arguments
- **Supervisor: Gagan Raj Gupta, Department of Computer Science and Engineering**



Projects

Volunteering

Agentic AI Workshop Series by Prof Raghav Kulkarni, CMI |
August 2026- Present

Conducting Sessions every Sunday
from 4PM to 7 PM on Agentic AI.

Coder's High Volunteer :
Volunteered for Workshop Series by
Sudarshan Iyenger, IIT Ropar

COVID-19 Response Volunteer:
Caregiver Saathi, May 2021

Hobbies

Room Recreation, Reading.

Others

.GATE DA 2025: Qualified
(March 2025)

Winner Hack Kochi #2:
(November 2019)
IEEE ORE, Kochi Hub

KEAM- Engineering: Rank 333
(August 2018)
Kerala State Entrance Examination

Highest Scorer: Science Stream:
Subject Topper for Maths, Chemistry and
Computer Science
(May 2018)

- Forecasting the Total Marine Fish Production In Kerala - Development & Comparative Analysis of Machine Learning Models**
Machine Learning, Fisheries | Jan 2022- May 2022
 - Marine Fish production Data in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from 1960 - 2014 .
 - Highest predictive power - Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07
- Chat Application, Publisher Subscriber Architecture | November 2019**
 - Co-Developed a real-time communication system using the MQTT protocol, applying a publisher-subscriber model to support chat-style applications as part of Hack-Kochi#2
- VR Industrial Training Simulator | Unity, C# | Jan 2021**

Co-developed a Virtual Reality application simulating boiler feedwater plant control room operations, creating a safe, immersive environment for industrial operator training. Collaborated on engineering interactive 3D scenarios and automation workflows in Unity/C#, effectively eliminating the physical risks and costs associated with traditional models.
- Prediction of Crime Categories | Ensembled Learning, Law & Order | August 2024**
 - The dataset consisted of 20,000 rows & 20 columns about location, victim demographics, date, time, etc. 6 crime categories were possible for the target variable.
 - Data Exploration, Cleaning and Preprocessing created 23 relevant feature columns.
 - LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** for which had an accuracy of 0.88080 in contrast to the baseline accuracy of 0.58
- Perspective Aware Sentiment Analysis| Applied Data Analytics | November 2025**
Models: Roberta-base, GPT-4o-mini
 - Real-time data pipeline using **Google News RSS** to fetch and parse headlines based on specific search queries and time windows.
 - Implemented a standard sentiment baseline using the **cardiff nlp/twitter-roberta-base-sentiment-latest** model to provide objective scores. Perspective-aware sentiment engine was created using GPT-4o-mini that adapts its analysis based on user-defined personas.
- Opacus - Differential Privacy in PyTorch**
Open Source Contribution, Data Privacy and Data Quality Project | May 2026
 - Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes. Worked on an issue that had a privacy bug (silent epsilon miscalculations with `WeightedRandomSampler`)
- LLM Distillation for Named Entity Recognition | Applied Machine Learning Project | May 2026**

Collaborated with a teammate to develop a knowledge distillation pipeline, compressing a 12B-parameter LLM (Gemma 3) into a 270M-parameter model for fine-grained NER.
- Dynamic Reflections: Probing Video Representations with Text Alignment | Introduction to Generative AI | June 2026**
 - Partnered with a teammate to evaluate and present cutting-edge research on multimodal video-text representation alignment and spatio-temporal semantics for the "Introduction to Generative Intelligence" course at CMI.

Education.

Chennai Mathematical Institute,
Data Science, Aug 2024 - May 2026

Master of Science